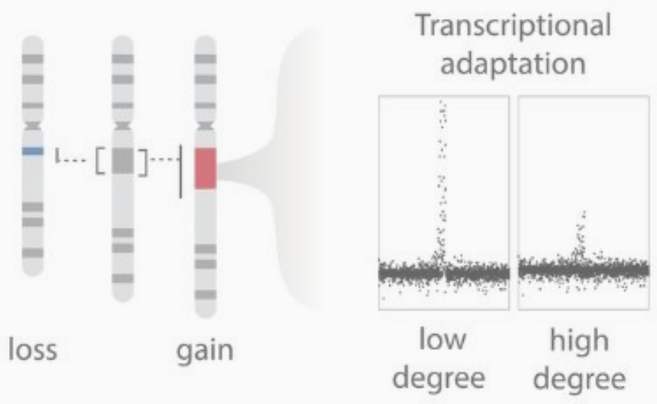
Spatial-TACNA: a novel methodology for unravelling the effects of CNAs in spatial transcriptomics

Feija Somefun[1], Marco de Bruyn[2,±], Arkajyoti Bhattacharya[1,±], Rudolf S.N. Fehrmann[1,±,*]

1 Department of Medical Oncology, University Medical Center Groningen, University of Groningen, Groningen, The Netherlands.
2 Department of Obstetrics and Gynaecology, University Medical Center Groningen, University of Groningen, Groningen, The Netherlands.
± Share senior authorship

Introduction

Copy number alterations



Clonal expansion



Intratumoral heterogeneity



Problem: without mapping the spatial distribution of CNAs, genetically distinct tumor populations and intratumoral heterogeneity are difficult to identify.

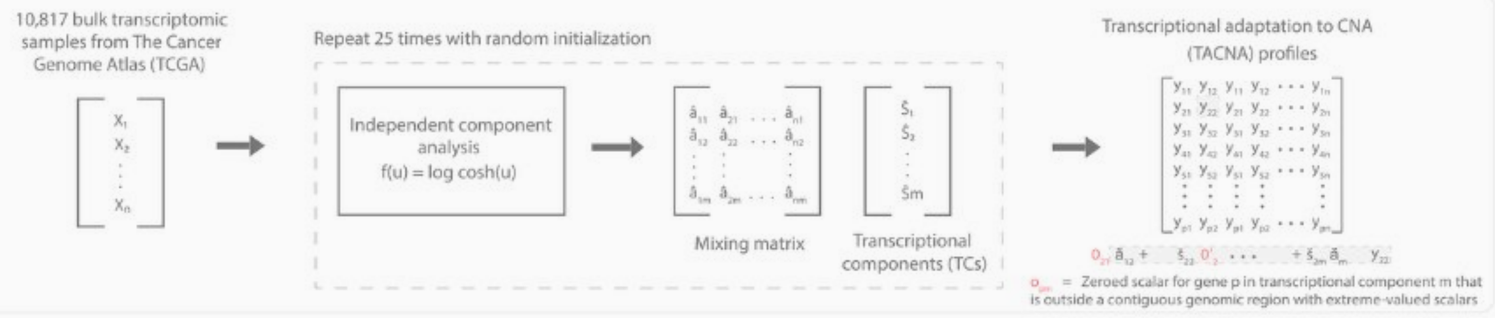
Aim

Spatial-TACNA aims to infer CNAs in spatial gene expression profiles while preserving local spatial variation.

Conclusion

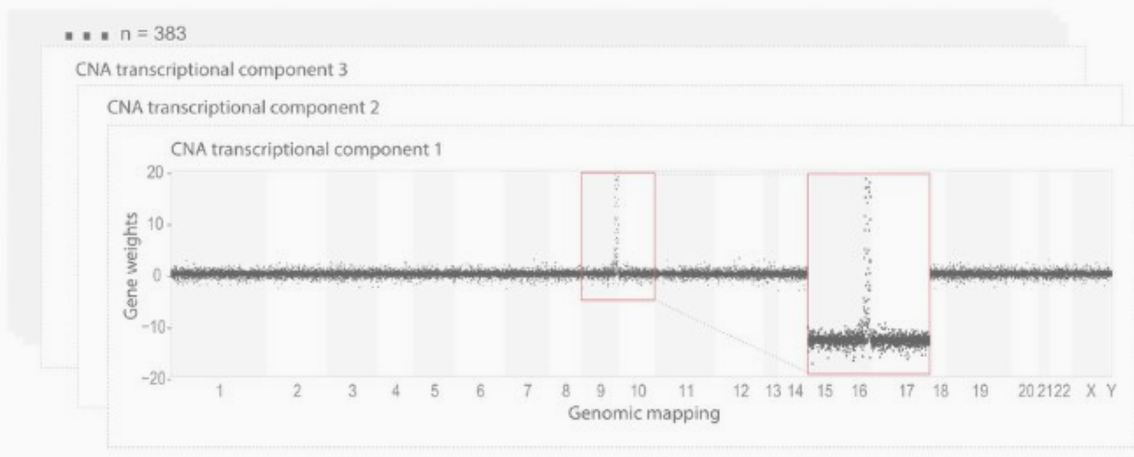
- Spatial-TACNA leveraged 10,817 bulk gene expression profiles to capture the effects of 383 CNAs (CNA-TCs). Utilizing these CNA-TCs to calculate their effect on gene expression, referred to as TACNA profiles, highly associated with SNP derived CNA profiles.
- Mapping these CNA-TCs enabled copy number inference at the level of individual spatial gene expression profiles for over 273 spatial transcriptomic samples. Spatial-TACNA revealed heterogeneous CNA burden across tumors, identified CNA-associated immune interactions, and linked spatial CNA patterns to functional pathways
- Spatial-TACNA provides a framework for investigating the spatial heterogeneity and transcriptional consequences of CNAs within tumors.

Methods



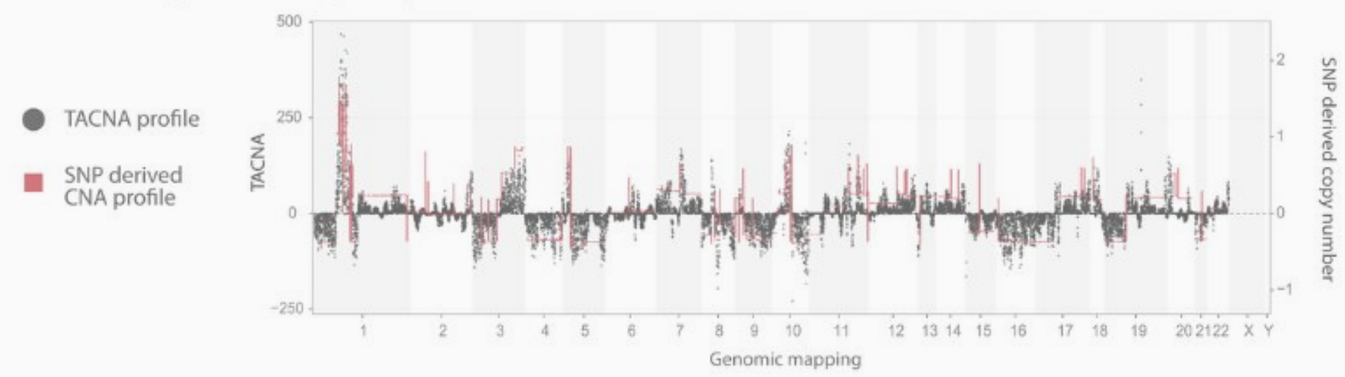
Results

383 transcriptional components capture the effects of CNAs (CNA-TCs)

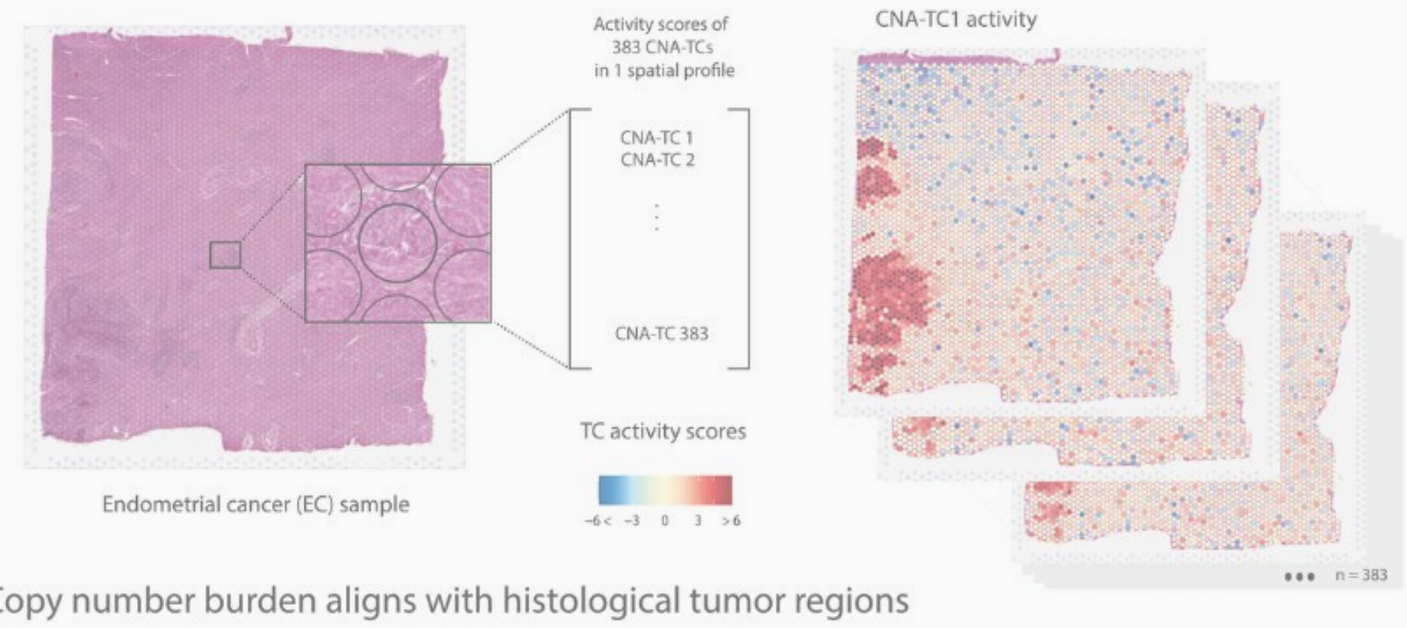


TACNA profiles are highly associated with SNP derived CNA profiles

Illustrated in a single bulk transcriptomic profile



Diverse effects of distinct CNAs identified in spatial gene expression profiles



Copy number burden aligns with histological tumor regions



unpublished

Supported by the MD/PhD Program of the Graduate School of Medical Sciences, University of Groningen and University Medical Center Groningen.